

Capstone Project Proposal

Name: [REDACTED]

Objective

As a life-long resident of Southeastern Connecticut, the submarine industrial complex is quite potent in my life. When interning for my districts congressman, I delved into the workings of our submarine industry and its influences on our communities. Therefore, for my capstone I would like to ask, how do Department of Defense submarine procurement contracts awarded to General Dynamics Electric Boat (EB) influence employment in Southeast Connecticut, and what can we predict about future workforce needs? Within this project, I can investigate how employment changes in New London County (EB's county) compares to other districts of the state and address if trends after procurement contracts have affects beyond SE Connecticut. Electric Boat uses materials and manufactured goods from local CT businesses in their submarines, so how stimulating is a procurement to the CT labor force as a whole?

Outcome

My project would use statistical analyses to link contract procurement to employment trends while also allowing me to create a forecasting model to help predict labor force demand. I can utilize visualizations of these analyses and forecasts to create an understandable deliverable for users of my project.

Impact

As a major industry of Connecticut, submarine procurement contracts are vital to the economic well-being of my district. Electric Boat provides jobs to skilled and unskilled workers and predicts it will continue large scale hiring up to 2033. One user could be the State of Connecticut to help them anticipate budget and infrastructure needs of an increased/decreased workforce over the years. Moreso, Electric Boat and their suppliers in CT can use this deliverable to help predict future demand, helping plan investments in employees and production. Schools (especially trade schools) around EB can also use this deliverable to estimate how their enrollment may change, or how their post-graduate hiring will change. Moreso, the EB workers union can use this data to predict how contracts may change within the company and to help plan strategic asks from higher ups to ensure workforce satisfaction and retention. In a broader lens, the general population of shipbuilders, engineers, and all other workers desired at EB can use this deliverable to help them determine best times to apply to open positions and local governments can use it to help predict economic impacts of increased workers and production in our towns.

Dataset(s)

[List your data sources with links to them. If you have already uploaded them to your capstone repository on GitHub, please mention the location. In addition, briefly discuss the datatypes and the reliability of the data.]

1. I collected a data set from the [Bureau of Labor Statistics Employment Projections](#). From this data I will extract occupations that are hired at EB. This is reliable data collected, organized, and regularly updated by government officials. This data set gives 2024 employment data and projected employment levels for each occupation for 2034. It also classifies level of education and work experience typically needed at entry level, as well as how much on-the-job training is needed.
2. I collected a data set on contracts with Electric Boat from the [Federal Procurement Data System](#) starting from 1995. Like the Bureau of Labor Statistics, the FPDS is upkept by federal employees, making it a reliable source. This data set gives the contract ID number, award type, date signed, info on the contracting agency, product and service code (ex: Submarine), and EB's information.
3. There is no publicly available data set of EB's hiring, so I had to use reporting from Connecticut's [Office of Military Affairs Annual Reviews](#) that reported EB's hiring numbers. However, they only started reporting it in 2019. So, I have hiring data from 2019-2024. Look in the challenges section for how I am attempting to overcome this.

Approach

1. First, I need to organize and clean my data into useful and understandable formats. This means separating out the occupations in the employment projections, ensuring I understand and can recognize specific terms and column names, and have a general organized csv file for each data set that I can easily refer to and work with
2. After this, I can start with my statistical analysis. I can write summary statistics and create general visualizations to help me observe trends. I can look at correlations and assess their importance. I can use techniques from my data mining class like clustering, classification, and principal component analysis to show hidden relationships, complementing traditional statistical modeling so I can reveal more non-obvious patterns in the data.
3. From this I can pick my relevant relationships between predictors to develop predictive models with traditional statistical methods, times-series analysis, and machine learning components so I can create a qualitative relationship between contracts and employment. I can also assess their effectiveness and predictive power.
4. At this point, I will be able to create an understandable deliverable for my users with user-friendly language and visuals
5. Then, I can write my final report and create my poster

Timeline

- (3 Weeks) Data Cleaning and Exploratory Statistical Analysis
 - Standardization of variable formats, join data, address NA values, etc.
 - Identify patterns, visualize trends, etc
- (3 Weeks) Feature Engineering and Modeling
 - Create and solidify predictor variables and their significance, develop regression and time series models to predict workforce demand.
- (3 Weeks) Modeling and Insights
 - Visualization of models for dashboard, Translate results into insights for policymakers, shipyards, and workforce planners.
- (1 Weeks) Compile into Dashboard/Deliverable
- (1 Week) Writing up the report
- (1 Week) Poster and Final Presentation

Possible Issues

1. There is no publicly available hiring data from EB, so I had to collect numbers from Connecticut's Office of Military Affairs annual reports and compile them myself. Reporting on hiring only started in 2019 in these reports. I have a family friend who was an executive at EB who is reaching out to connections at the company to see if there is any data available.
2. I think I will have to spend a large chunk of time sorting through the data and creating an understandable format for all of them. While I have been exposed to a lot of the terminology and acronyms used in the industry, there is still a lot for me to learn and familiarize myself with. I am planning to write myself a glossary of sorts to ensure I have a key to look to, so I have consistent understandings over the course of the semester of terms.
3. There is also probably a lag between contract signing date and when a change in employment can be observed. Attempting to work around or discover this lag may be a challenge. I will probably have to play around with different lags and see what fits.